

Is an Engineering Degree Safe From AI? The 2026 Risk Score

The free sampler — the score by track, the six factors in brief, and the honest reasons to think twice. The complete profile goes deeper.

3.4

AI EXPOSURE · CIVIL / STRUCTURAL (PE, SITE-BASED)
where 10 is most at risk

Re-scored every six months. We publish where we might be wrong.

Quick answer: Engineering scores between **3.4 and 6.3 out of 10** for AI exposure depending on the discipline, where 10 is most at risk. Civil and structural engineering with a PE license and site work scores 3.4. Desk-based design and analysis scores 6.3. Engineering is the instructive contrast to computer science — the same technical aptitude, pointed at the physical world, picks up three protections software does not have.

The short answer for parents: yes — and for a technically-minded student, the engineering-versus-computer-science comparison deserves far more thought than it usually gets. Software pays more at entry, where entry survives. Engineering holds licensure, site work, and legal accountability for things that must not fail.

Engineering AI risk score by discipline

Every career in this index is scored 1–10, where **10 is most exposed** to AI. Same six factors, same weights, applied identically to an engineer and a paramedic.

DISCIPLINE	2023	2025	FALL 2026	3-YR MOVE	BAND
Civil / structural (PE, site-based)	3.4	3.7	4.0	+0.6	Low–Moderate
Electrical power & grid	3.3	3.6	3.9	+0.6	Low–Moderate
Mechanical (industrial)	3.9	4.1	4.4	+0.5	Moderate
Manufacturing / process	4.4	4.7	4.9	+0.5	Moderate
Design & analysis (desk, CAD-centered)	5.2	5.8	6.3	+1.1	Moderate–High

2023 and 2025 figures are reconstructed using current methodology, not archived from past editions.

The comparison that matters most: entry-level software development scores **8.1**, senior software architecture **5.4**. Licensed site engineering scores **4.0**.

Engineering vs computer science: the comparison worth making

Same kind of mind. Same math. Very different exposure — and the reason is structural rather than technical.

	SOFTWARE	ENGINEERING
Regulatory protection	None — no license exists	Strong — the PE stamp is a legal monopoly
Physical requirement	None — the work is the medium of the threat	Real — sites, commissioning, inspection
Accountability for catastrophic failure	Commercial	Legal and personal
Entry route	Purely market-driven, and narrowing sharply	Licensure requires supervised experience years
Entry-level score	8.1	4.0 licensed / 6.3 desk

Software has two of our six protection factors sitting at essentially zero for everyone in the profession. Engineering has neither of those gaps at the licensed end.

That is the single most decision-relevant comparison in this index for a student who is good at math and likes building things.

Will AI replace engineers?

Not the ones who sign, visit sites, or make calls where failure is catastrophic. It is taking a substantial share of the desk layer.

AI IS TAKING	IT CAN'T TOUCH
CAD drafting and documentation	The PE stamp and the liability behind it
Standard-case calculation	Site work, commissioning, inspection
Simulation setup and iteration	Judgment on one-off systems with thin precedent
Generative design option-space search	Deciding which option is actually buildable here
Compliance and code checking	Being accountable when a structure fails

Generative design tools now produce and compare many options against constraints of cost, material and structure. Each of those tasks was, until recently, a junior engineer's assignment.

Why does licensed engineering score low? The six factors

How civil and structural engineering rates against each. Ratings are 0–10 on each factor's own terms.

FACTOR	RATING	EFFECT ON RISK
How much of this job can AI already do?	6.5	↑ moderate exposure
How hard will it be to land that first job?	3.0	↓ good protection
Does it have to be done in person, with your hands?	6.0	↓ moderate protection
Does someone need a human they can trust and hold responsible?	7.5	↓ good protection
Does the law require a licensed human?	9.0	↓ strong protection
How often does the job hit genuinely new, high-stakes situations?	8.0	↓ strong protection

ONE FACTOR, WORKED THROUGH IN FULL

So you can see what the analysis actually looks like.

How often does the job hit genuinely new, high-stakes situations? — rated 8.0

AI is excellent at patterns and weak at genuine novelty. So this factor measures a specific thing: how much of the work involves situations that do not match anything the system has seen,

where being wrong is expensive.

Engineering rates high, but not for the reason most people assume. It is not that the calculations are hard — calculations are exactly what machines do well, and standard-case analysis is among the fastest-automating parts of the profession.

It is that **every site is different, and the consequences of being wrong are structural**. A bridge sits on particular ground with particular water, in a particular seismic and thermal environment, subject to a particular jurisdiction's code, built by a particular contractor with particular constraints. The combination has no precedent even where every individual element does.

That is why the licensed, site-based track holds at 4.0 while desk-based analysis has moved to 6.3. The analysis is generalizable. The judgment about whether the analysis applies *here* is not.

One honest caution, and it is the profession's real problem. The PE license requires supervised experience years, which keeps the on-ramp legally open — the same mechanism that protects nursing and medicine. But if those years are increasingly spent supervising generated output rather than doing the analysis by hand, the question becomes whether they still build the judgment the license is meant to certify. The rung exists. What it teaches is changing, and the profession has not really debated this yet.

Which engineering disciplines are safest from AI?

Ranked by exposure, safest first:

1. **Electrical power and grid — 3.9.** Physical infrastructure, licensed, and riding the largest demand wave in the sector. 2. **Civil / structural — 4.0.** The stamp, the site, and one-off conditions on every project. 3. **Mechanical (industrial) — 4.4.** Physical systems in complex environments, with more standardized design content. 4. **Manufacturing / process — 4.9.** Structured environments, which is where automation advances fastest. 5. **Design and analysis (desk) — 6.3.** The fastest-moving track, and the one most exposed to generative tools.

Electrical engineering pointed at power and infrastructure may be the single most demand-secure choice in this entire index. The AI buildout is itself the demand — data centers, grid capacity, electrification.

Common questions

Will AI replace civil engineers? Not the licensed, site-based role. The PE stamp is a legal monopoly and site judgment is unpredictable physical work. Desk-based analysis is a different story and is moving fast.

Is engineering safer than computer science? At entry level, considerably — 4.0 for licensed engineering against 8.1 for entry-level software development. Software pays more where entry-level hiring survives, and engineering carries protections software does not have at all.

Which engineering discipline has the best future? Electrical, on current evidence, particularly in power and grid work. The infrastructure demand created by AI itself is the largest sustained tailwind in the sector.

Do I need a PE license? For the protection this index measures, yes — it is the difference between the licensed track at 4.0 and the desk track at 6.3. It also mandates the supervised experience that keeps the entry route open.

Is CAD work being automated? Substantially, and it is among the faster-moving tracks we score. A degree used as a generic analysis credential at a desk is the exposed version of engineering.

Related careers

- **Computer science** — 8.1 entry-level, 5.4 senior. Free to read in full.
- **Skilled trades** — 2.5 field and service, 5.0 structured settings
- **Architecture** — 5.0 licensed, 7.6 production
- **Construction & project management** — coming in this edition
- **Data science & analytics** — coming in this edition

What's in the engineering Career Value Guide

This sampler tells you where engineering stands. The full Career Value Guide tells you what to do about it.

	SAMPLER	CAREER VALUE GUIDE
Verdict, all discipline scores, 3-year trend	✓	✓
Six-factor ratings	✓	✓
Reasoning behind every factor rating	one example	✓
How durable each protection is — where generative design is heading		✓
The full software-versus-engineering comparison		✓
The honest downsides — pay relative to software, licensure timelines		✓
What's genuinely good about it — and who this work suits		✓
The AI-native advantage — how to prepare, and why this cohort has an opening		✓
Routes in — degree, apprenticeship-to-degree, and the PE pathway		✓
Where it leads later — including infrastructure and energy		✓
Program evaluation checklist — PE preparation, co-op placement, field exposure		✓
Questions to ask a program — plus red flags		✓
Sourced further reading		✓
Discussion questions for parent and student		✓
A short version written directly to the student		✓
Technical scoring appendix		✓

GET THE CAREER VALUE GUIDE

Most families are weighing two or three careers seriously, and a few more they haven't ruled out. Pick the ones you need.

1 career	\$49	Choose →
3 careers	\$69	Choose →
Unlimited	\$99	Choose →

Each Career Value Guide includes everything in the table above, plus a short version written directly to the student and the technical scoring appendix. This edition and the next are included — we re-score every six months, so what you buy stays current for a year.

[Read a complete Career Value Guide free →](#) We publish one Career Value Guide in full, openly, so you can judge the depth before buying anything. Computer science is the one to read — it's the most surprising result in the index.

Pivotum scores thirty careers against AI exposure using a fixed, published methodology, re-scored every six months. Scores measure exposure to what AI can already do — not how much any particular employer has deployed. We publish where we might be wrong.